

8. [Maximum mark: 8]

(a) Show that $\sin 3\theta \equiv 3\sin \theta - 4\sin^3 \theta$. [4]

(b) Hence, describe a sequence of transformations that maps the graph of $y = \sin \theta$ onto the graph of $y = 6\sin\left(\theta + \frac{\pi}{6}\right) - 8\sin^3\left(\theta + \frac{\pi}{6}\right)$. [4]

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